

Statistics-basics (and some advanced stuff in brief)

- Introduction for experimental and clinical research
- Andreas Busjahn

Chance / Probability / Uncertainty



Statistics: Structure / Count / Hide (?)

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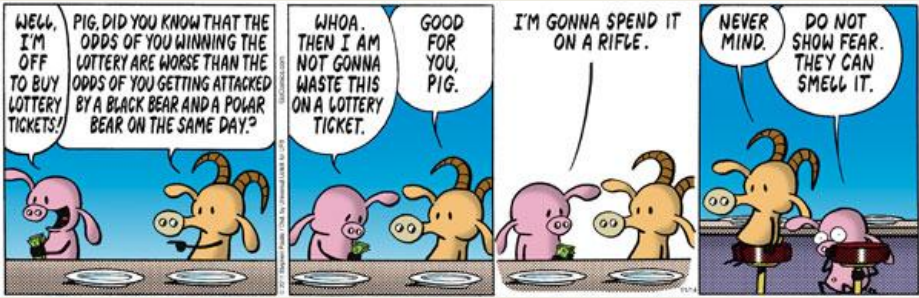


Pictures: Urs Wehrli

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Probabilities in Daily Life

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How to Get Probability:

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- Probability Lotto?
 - Computed,
 $1/(10.068.347.520/720) = 1/13.983.816$
 0.000007151124%
 - Empirical
- Probability bear attack??
 - Theoretical?
 - Empirical
- Meaningful?? Relevant??

$$= \frac{1}{\frac{49!}{6! \cdot 43!}}$$

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Real World Statistics

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Survivorship Bias

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- Learn from the best?
- Statistician Abraham Wald was asked in WWII to compute optimal patterns to enforce airplanes
- Base: hit patterns of returning planes
 - **Idea: reinforce areas with highest hit probability**
 - **Wrong!**
 - **Never or seldomly hit areas need protection**
 - **Why?**
 - **Don't imitate the successful, but avoid errors of those that fail...**

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Statistical literacy

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- 100% risk increase for thrombosis due to new generation of birth control pill ->
 - Panic, discontinued use
 - 13,000 additional abortions in UK
 - Real risk: from 1 in 7,000 to 2 in 7,000
- Face recognition with AI to detect suspects: Trial in Germany
 - Reported as success (but never implemented)
 - 80% of volunteers playing suspects are detected
 - Only 0.1% false positives
 - 12 million daily passengers -> 12,000 false positives

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Workflow...

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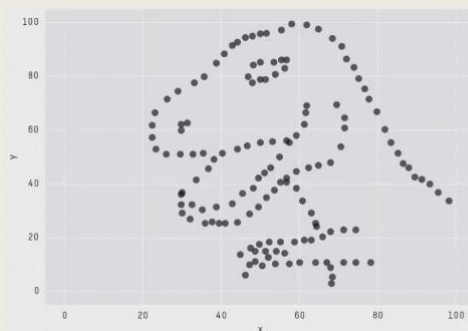
- Statistical analysis plan (hardly ever, rough idea)
- Import / cleaning
- Scale level / data type (may change)
- Graphical exploration
- Distribution check / decision
- Descriptive statistics
- Statistical modelling / testing
- Reporting

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Same Stats Different Data

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What is hidden in your data? (?Datasurus?)



X Mean: 54.2659224
 Y Mean: 47.8313999
 X SD : 16.7649829
 Y SD : 26.9342120
 Corr. : -0.0642526

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Data Structure

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ID	Group1	Group2	sysRR pre	diaRR pre	sysRR post	...
A1	Cntrl	WT				
A2	Cntrl	WT				
B5	Treatm	TG				
...						

• Structure depends on software and goals

• Basic structure:

– First row variable names

– 1 row 1 subject

– No computation in rawdata sheets!

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Typical Data Mess

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	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	
1	RETROdatabase traditionelle Risikofaktoren vs cTnI als Prädiktor zukünftiger Ereignisse (n=8)																	
2	FU-revasc(n=8) vs noFU-revascALL bzw vs noFU-revascCAD. Dann FU-AP/ST vs FU-noAP/ST																	
3	traditionelle Risikofaktoren sex (m=1,w=0), age(>60=1, bis 60=0), Hypertonus (0 vs 1), Nikotin (0vs1), HLP (LDL>130 (3xohne LDL bei Chol200)bei LDL																	
4	baseline																	
5	Pat-N	INA-N	GE	Vorname	sex	age	age	m	kg	Grö²	BMI	BMI	AMI	EF [%]	EDD [mm]	Begleit-Ds	dia	
6					m=1	>60=1					>30=1					a-numerisch		
7																		
8	1101	###	1	Hans-Dieter	1	1	63	1,78	75	3,168	23,7		0	65	54	HypI, HLP		
9	1102	###	1	Ursula	1	1	66	1,68	77	2,822	27,3		0	60	44	HypI, Adip, HLP, Nik		
10	1103	###	2	Klaus	1	1	71	1,76	72	3,098	23,2		0	65		HLP		
11	1104	###	1	Bernd	1	1	52	1,69	88	2,856	30,8	1	0(laevo60)	60	47	nikot, HLP		
12	1105	###	1	Gerhard	1	1	65	1,86	103	3,46	29,8		0	EFlaevo65,	60	52	HypI, HLP	
13	1106	###	1	Herbert	1	1	72	1,7	70	2,89	24,2		1(VWV197)laev	50	47	Hyp, HLP, ASI, AII		
14	1107	###	1	Günter	1	1	73	1,74	72	3,028	23,8		0	65	50	HLP		
15	1108	###	1	Orthild	1	1	58	1,62	96	2,624	36,6	1	1(HW186)(lae	40	41	Hyp, HLP, Diablib		
16	1110	###	1	Meryem			52	1,55	60	2,403	25		0	75	39	HypI		
17	1111	###	1	Gerd	1	1	52	1,76	85	3,098	27,4		0	60	48	HypI, HLP		
18	1112	###	1	Alfons	1	1	66	1,67	70	2,789	25,1		0	60	48	HypI, HLP		
19	1113	###	1	Jürgen	1	1	65	1,74	86	3,028	28,4		0	60	61	Hyp, Adi, TIA, LSB Schoenesh		
20	1114	###	1	Gerd	1	1	61	1,78	93	3,168	29,4		0	65	54	Hyp2, Adip, HLP Neuk		
21	1115	###	1	Wolfgang	1	1	60	1,75	96	3,063	31,3	1	0	47	43	HypIDDM, HLP, Adip, AVK-Znl		

▪ Several rows with headline/variable names

▪ Empty cells with a meaning (" " instead of „w“) (... Missing Value?)

▪ Numeric data and text within a variable

▪ Entries with varying delimiters within cells

▪ Synonyms (medication, categories...)

▪ Color-coding / font family for selection (... usually lost during import, but for tidyxl)

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Why test?

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- If you bet on a flipping coin several times and get head over and over, when do get suspicious that your partner is cheating? After 2? 3? 4? 5? 10?
- A series of 4 is still not too unlikely (6.25%), but a series of 5 has a probability of 3.12% only
- Treating a patient with a new drug and coding response is no different from flipping a coin...
- So how many patients have to benefit from a drug before you'd say this is worth increased hc spending?
- If you played for hours already (or conducted enough experiments) , such a 'rare' event becomes more likely
- Statistical tests help you figuring out if something could just be due to chance

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Population vs. Sample or Why Statistics at all?

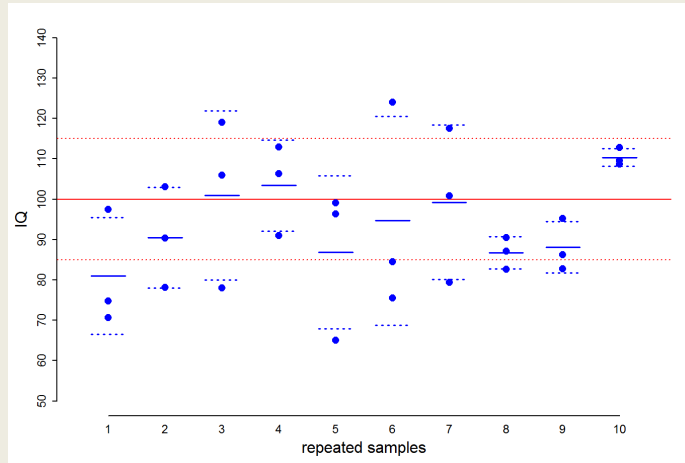
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- You never can measure a population, so you take a (random) sample
- Accordingly you may never know the 'real' descriptors of the population, but you make an estimation based on your sample
- If you'd do this again, you'd get different estimates but those differences are just random
- Statistical tests standardize estimation and decision

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Random Samples (no systematic difference!)

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Statistics is but a tool (careful, a sharp one)

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We have the duty of formulating, of summarizing, and of communicating our conclusions, in intelligible form, in recognition of the right of other free minds to utilize them in making their own decisions.

R.A. Fisher 1955

Garbage in garbage out:

Mathematics may be compared to a mill of exquisite workmanship which grinds you stuff of any degree of fineness, but nevertheless what you get out depends on what you put in - and as the grandest mill in the world will not extract wheat flour from peasecuds so pages of formulae will not get a definite result out of loose data.

TH Huxley

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Why YOU have to THINK

NESTLÉ & L'ORÉAL RESEARCHERS INVENT THE FUTURE OF MEN'S CARE.

innéov

A new generation of biactive phytoactive concentrates. 20 patents pending.

Efficacy measured in clinical studies under dermatological control.

innéov man ANTI-HAIRLOSS

Every hair has a growth motor at its base: the capillary plexus. When healthy it generates new cells and hair fibers anchored, thick and resistant. But the capillary plexus can weaken under the influence of 4 main stresses:

- inflammation
- bulb stress
- slow down in microcirculation and provision of nutrients
- oxidative stress: free radicals, principally present in the temples and crown area of a man's head.

Deprived of its motor, the hair miniaturizes and falls prematurely.

Empowered by their dual expertise in nutrition and in hair biology, innéov scientists create:

innéov man ANTI-HAIRLOSS

The 14 biactives + strength conditioners + with Phytosterols & Taurine which acts at the very heart of the bulb, to continue its growth.

INNOVATION:

For the very 1st time in an anti-hairloss nutritional concentrate for men, innéov scientists, directed by the innéov scientists, have found a concentrate in phytoactive (99.9%), this unique extract has been developed with a taurine complex (taurine) according to light from within against attacks on the capillary plexus involved in men's hair loss.

EFFICACY measured in clinical studies under dermatological control:

innéov man ANTI-HAIRLOSS benefits from a clinical study* carried out on 102 subjects during 16 weeks with a daily intake of 2 capsules, following a strict and rigorous methodology.

The measured clinical results concern:

- **slowing down of the hairloss process** (decrease in the number of hairs in loss phase in certain precise zones, such as the edge of the temples).
- **capillary density**, which relates to the number of hairs anchored by cm².
- **quality of hair** and more particularly hair strength and thickness.

From 2 months, men notice the results**:

- 76% say the product fights effectively against hairloss and increases hair density.
- 83 % acknowledge their hair is stronger, its quality has improved.

* Clinical study EC:CAP-2007-02
** Consumer study carried out on 123 men

Answers to all your questions at www.inneov.com

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You'd better have a clear hypothesis!

IT SEEMS THERE IS A DIFFERENCE BETWEEN A "HYPOTHESIS" AND A "GUESSTIMATE!"

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Significance and Power

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- Working hypothesis: This is what you expect! E.g. treatment is lowering blood pressure more than placebo, transgenic animals become obese...
- Null hypothesis: This is what you test! No difference / relation, BP under therapy = BP under placebo
- 4 possibilities:
 - ▶ Null hypothesis correct, test false positive (case A): alpha-error
 - ▶ Null hypothesis correct, test correct negative (case B)
 - ▶ Null hypothesis false, test false negative (case C): beta-error
 - ▶ Null hypothesis false, test correct positive (case D)
- Significance: NOT probability of case A, but probability of your data given the NULL hypothesis, calculated from your data, conventionally <0.05
- Power: Probability of case D, 'estimated' based on assumptions about effects and sample size, 'calculation' would require knowledge of true difference, conventionally set at 0.80

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Wrong measures -> wrong results

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Samplesize

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The Question of Sample Size

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- How many subjects does it take to have a fair chance to find the expected effect?
 - What is known about variability of measure of interest?
 - Is variability expected to be the same in all groups?
 - What exactly is the effect you want to prove (take the smallest relevant rather than your best hope!)
 - What chance do you want to have to find it (80%)
 - (Have someone) Run the appropriate calculations
- What is the smallest effect that can be detected with a given sample size?
- What is my chance (power) to find the expected effect with a given sample size?

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Example Sample Size Estimation in R

New Treatment -> delta EF

Power=0.8, p= 0.05 , one.sided , paired

population 1 with inclusion criteria:
2%< AF burden >95%

Recurrence placebo: 0.5 Recurrence placebo: 0.6 Recurrence placebo: 0.7 Recurrence placebo: 0.8 Recurrence placebo: 0.85

Sample size

reduction

recurrence rate verum change vs. placebo

0 -0.05 -0.1 -0.2 -0.3

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Know your data!

- Boxplot
 - Box: Median, 25%, 75%
 - Whiskers: Extreme values (except outliers)
 - Outliers: Data outside box $\pm 1.5 \cdot \text{Interquartile range IQR}$

body_mass_g

species

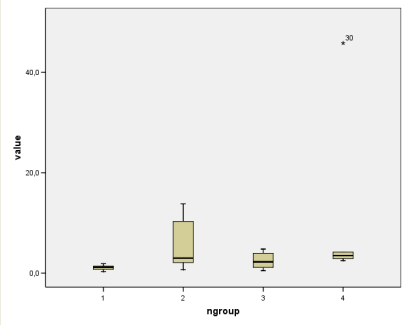
Adelie Chinstrap Gentoo

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Outlier

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- Sometimes one or a few values are far away from others
- 1st check for data entry or experimental problems (“that tube looked funny anyway”)
- Outlier can be caused by biological variability (rare extreme) or undetected experimental problems (bad pipetting, voltage spike, filter hole...)



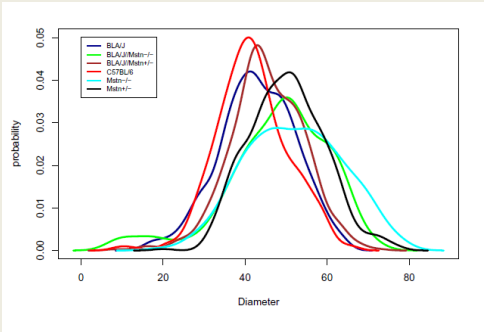
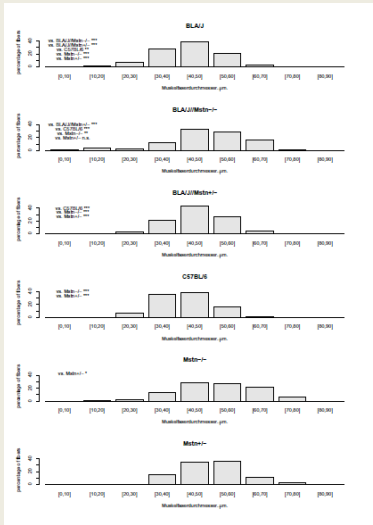
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Graphical Display of Data

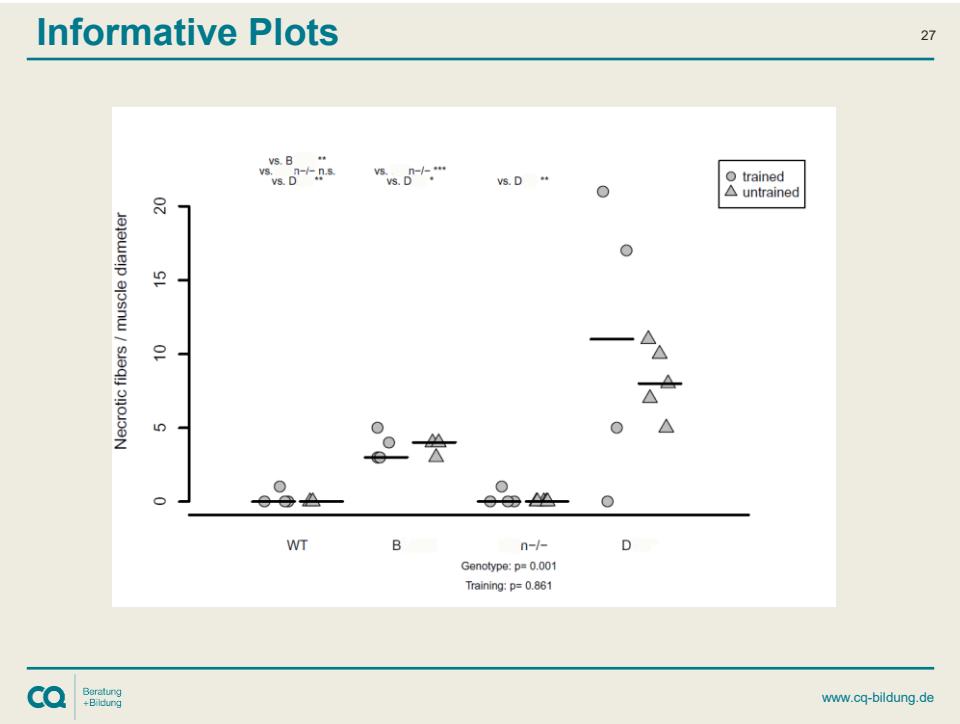
26



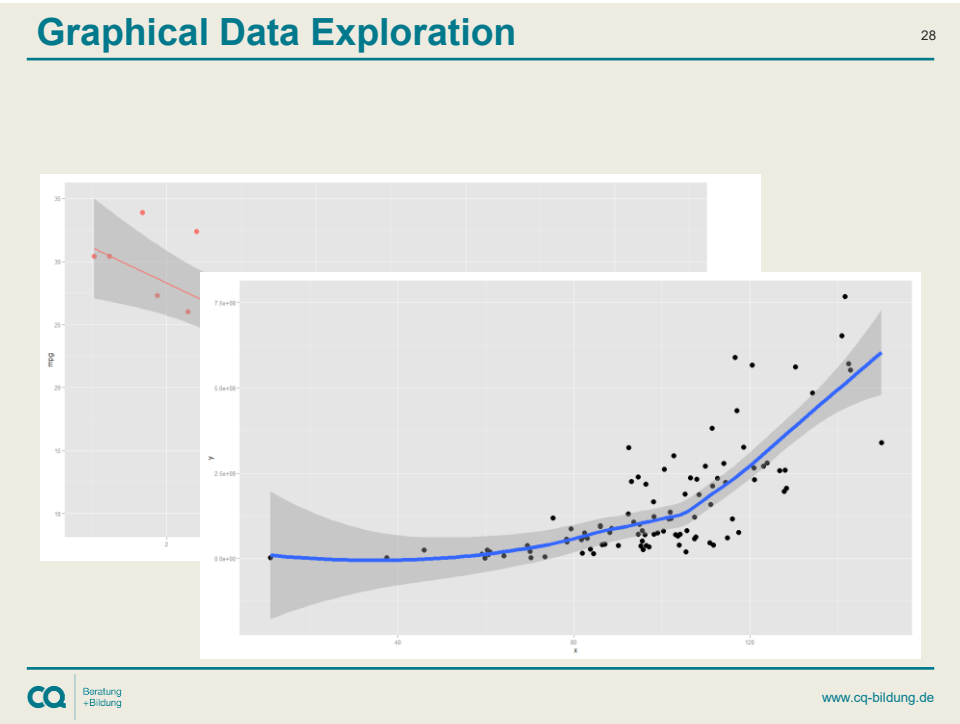
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Classification Levels

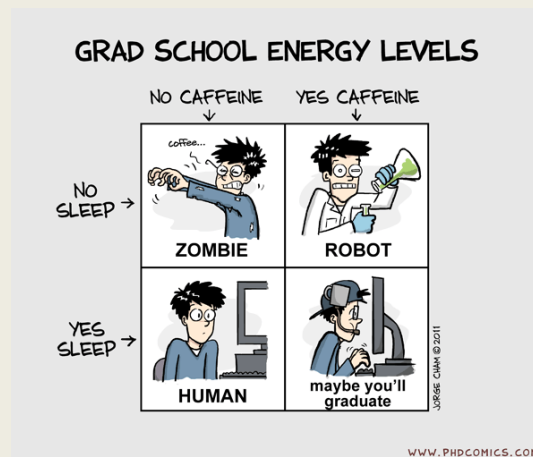
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- **Discrete (→nonparametric tests)**
 - Nominal measurement (just different)
 - PLZ/ZIP-Code, sex, medication, rat-strains...
 - Ordinal scale (defined by order)
 - (ordered categories)
 - Measurements without equidistant scale or Normal distribution
 - Percentages (at least in very low/high range)
- Ordinal Categories
 - NYHA, disease level, school/university grades
- Ordinal Measures
 - Measurements without equidistant scale or Normal distribution
 - Percentages (at least in very low/high range)
- **Continuous (→parametric tests)**
 - Interval scale (defined by constant distance)
 - temperature in °C, IQ, many Bioassays
 - Ratio scale (absolute zero)
 - temperature in K, blood pressure, most biologic measures

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Cross-Tables

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Cross-Table / χ^2

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Risk * PatientGroup Crosstabulation

		PatientGroup		Total
Count		1,00	2,00	
Risk	no	25	29	54
	yes	23	23	46
Total		48	52	100

- 2 classifications, e.g. screening positive / disease positive
- Test for random distribution across table cells
- Computation of difference between empirical and expected number per cell
- Expected: $\text{row} \times \text{column} / \text{total}$
- If expected n in at least one cell < 5, then Fishers Exact Test
- χ^2 : 0,137 / 1df, p=0,71
- Both patient groups have equal risks (no relation between proposed risk factor and disease)

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Computation χ^2

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```

> chi_out$observed
  [,1] [,2]
[1,] 25 29
[2,] 23 23
> chi_out$expected
  [,1] [,2]
[1,] 25.92 28.08
[2,] 22.08 23.92
> chi_out
Pearson's Chi-squared test
data: crosstable
X-squared = 0.13651, df = 1, p-value =
0.7118

> chi_out$observed-
  chi_out$expected
  [,1] [,2]
[1,] -0.92 0.92
[2,] 0.92 -0.92

> .92^2/25.92+.92^2/28.08+
  .92^2/22.08+.92^2/23.92
[1] 0.1365147

```

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Cross Table in R

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- `table(..., useNA = c("no", "ifany", "always"))`
- `table(rawdata$geschlecht, rawdata$testmedikation)`

	0	1
m	12	12
w	2	2
- `chiOut<-chisq.test(geschlecht, NYHA)`
 Pearson's Chi-squared test
 data: rawdata\$geschlecht and rawdata\$NYHA
 X-squared = 6.4278, df = 3, p-value = 0.09255
- `chiOut$p.value`
 [1] 0.09255
- `chiOut$expected`

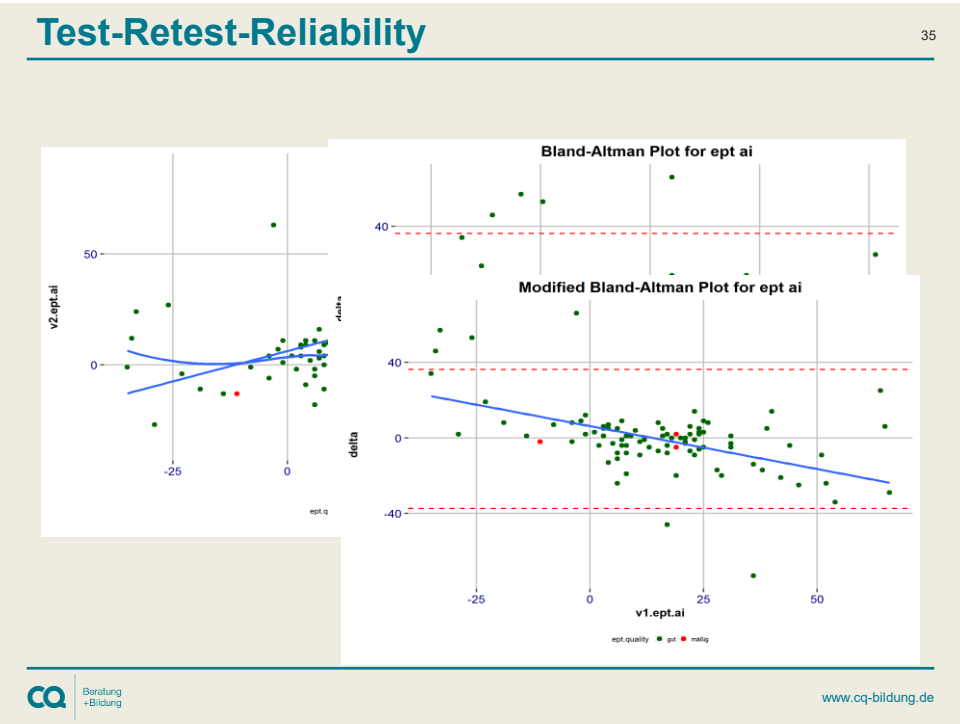
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Characteristics of a Diagnostic Test 1

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- Validity:
 - Degree to which a test measures what it was designed to measure
 - Tested against a gold standard (what is it, what if new test is better?)
- Reliability:
 - Consistency of a set of measurements or measuring instrument
 - Do measurements of the same instrument (test-retest) give same measurement
 - Do two independent assessors give similar scores (inter-rater reliability)
 - Reliability does not imply validity, a reliable measure is measuring something consistently, but not necessarily what it is supposed to be measuring

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Results of a Diagnostic Test

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		Trait (Disease/Pregnancy...)	
		Present +	Absent -
Test	Positive T+	Correct positive a	False positive b
	Negative T-	False negative c	Correct negative d

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Characteristics of a Diagnostic Test 2

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- Sensitivity:
 - Probability that test is positive in case of disease
 - $P(\text{positive test} \mid \text{disease+}) = a / (a + c)$
- Specificity:
 - Probability that test is negative in case of no disease
 - $P(\text{negative test} \mid \text{disease-}) = d / (b + d)$
- Positive prognostic value:
 - Probability that someone with a positive test has disease
 - $P(\text{disease+} \mid \text{positive test}) = a / (a + b)$
- Negative prognostic value:
 - Probability that someone with a negative test has no disease
 - $P(\text{disease-} \mid \text{negative test}) = d / (c + d)$

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Buy company share? Do Testkaufen?

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- Stock exchange news: Epigenomics confirms value of Screening (*4. April 2006*)
- Epigenomics AG, a biodiagnostics enterprise, documented by new data that free DNA in blood is a key for early detection of colon cancer. As reported during the 97. annual AACR meeting in Washington, methylated DNA for the Septin9-gene is found in plasma of *up to 57 % of* all patients with colon cancer of various stages, with a high *specificity (95%)*. "Methylated Septin9-DNA in blood is a marker outstandingly suitable for Screening of large populations for asymptomatic colon cancer ..."
- Sollte man Aktien kaufen? (Bzw. den Test kaufen und anwenden?)

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Test Colon Cancer

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		Disease	
		Present	Not present
Test	T+	30	47
	T-	30	893

- Incidence: 6%
- Specificity: 95%
- Sensitivity: 50%
- Pos. progn. value: 40%
- Neg. progn. value: 97%
- (without test 95%)

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Trading Range

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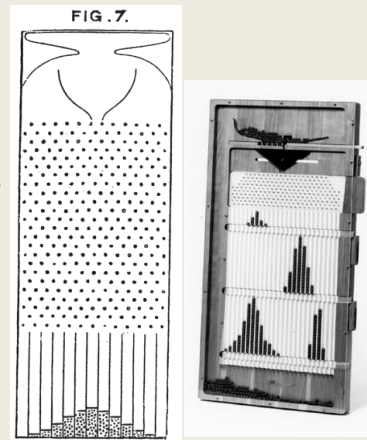
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Why 'Normal Distribution' is Normal

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$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Galtons Quincunx (or Bean Machine)
 - ▶ Device invented by Sir Francis Galton to demonstrate the law of error and the normal distribution
 - ▶ Vertical board with interleaved rows of pins. Balls are dropped from the top, and bounce randomly left and right as they hit the pins. Eventually, they are collected into one-ball-wide bins at the bottom. The height of ball columns in the bins approximates a bell curve.
 - ▶ Think of genes instead of pins...

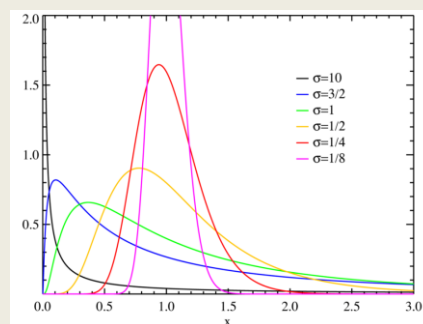


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Log-normal Distribution

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- If factors do not act additively but multiplicatively
- Frequent in biology and economy
- Even letters/word in daily English!

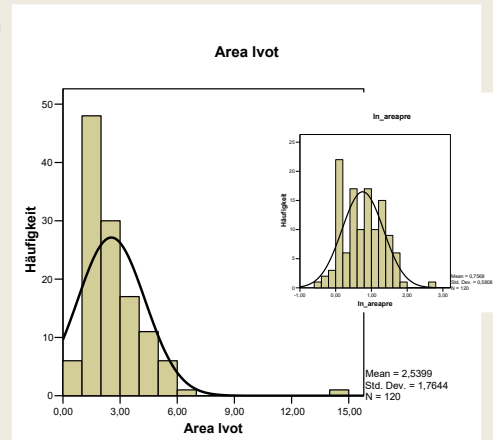


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Test for Normal Distribution

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- Many tests require continuous data from a Normal distribution in the population
- Kolmogorov-Smirnov-Test compares empirical and theoretical distribution (Normal, Poisson, Uniform, Exponential)
- Sometimes, data can be transformed to Normality (e.g. skewed distributions), log, root, ...
- Many tests are somewhat robust against deviations; decision between non-parametric test vs. transformation is difficult



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Distribution Test in R

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- `ksOut<- ks.test(rawdata$groesse.screening, 'pnorm', mean(rawdata$groesse.screening,na.rm=T), sd(rawdata$groesse.screening,na.rm=T), exact=F)`
- `ksOut$p.value`
- `shapiro.test(rawdata$groesse.screening)$p.value`

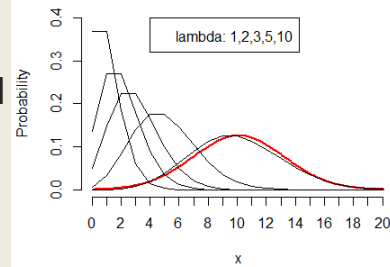
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Other distributions: Poisson

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- For count data (e.g. number of plaques in a region, no. drugs...)
- Discrete, no negative values
- Asymmetric, skewed to the right
- Mean=Var: lambda
- For large lambda-> Normal distribution

$$p(x) = \frac{\lambda^x}{x!e^\lambda}$$

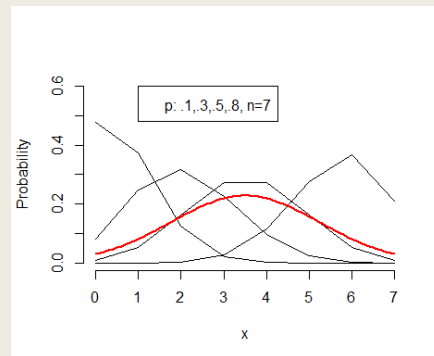


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Other distributions: Binomial

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- For yes/no events (e.g. responder, rehospitalization)
- Out of n observations, what is the probability of having 0,1,2,...n events?
- Discrete, no negative values
- Asymmetric (except for p=.5)
- Mean=n*p, Var=n*p*(1-p)
- For p=.5 and large n-> Normal distribution



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Standard Deviation / Standard Error

• Sample n:	100	10	10
• Mean:	100	100	100
• Variance:	100	100	400
• Standard deviation			
SD:	10	10	20
• 68% of all values within mean ± SD			
• Coefficient of Variation			
CV:	10%	10%	20%
• Standard error of the mean			
SEM:	1	3	6

Sum (x) / n

Sum ((x-mean)²)/(n - 1)

Square root (Variance)

SD/mean*100%

SD/Square root (n)

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Other ‘Means’ and Measures of Variation

- Geometric mean:
 - Bacteria growth, mean of relations $\sqrt[n]{x_1 * x_2 * ... * x_n}$
- Mode:
 - Category with highest frequency (1 1 2 3 4 4 **6 6** 8 11)
 - Peak of distribution
- Median:
 - Value that divides upper/lower 50% of sample (1 1 2 3 4 **4** 6 6 6 8 11)
- Range
 - Useful for $n \leq 13$ $x_{max} - x_{min}$ (11-1=10)
- Percentiles (usually 25%/75%) 2,5 / 6

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Descriptives in R

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- `mean(rawdata$groesse, na.rm=T)`
- `round(mean(rawdata$groesse,na.rm=T),2)`
- `sd(rawdata$groesse,na.rm=T)`
- `#SEM (library wrappedtools)`
- `median(rawdata$groesse,na.rm=T)`
- `quantile(rawdata$groesse,probs=c(.05,.95), na.rm=T)`

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Confidence Interval

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- **Mean $\pm$ SEM includes true mean with 68% probability**
- **In general CI is given by**

$$mean \pm z * \frac{s}{\sqrt{n}}$$

- **Quite meaningful is 95% CI:**
 - $Z=1.96$ (mean ± 1.96 SEM)
- **For small samples ($\approx n < 100$) t instead of z**
- **Higher security is paid for by less precision, precise (narrow interval) means less certain**
- **CI for proportions**

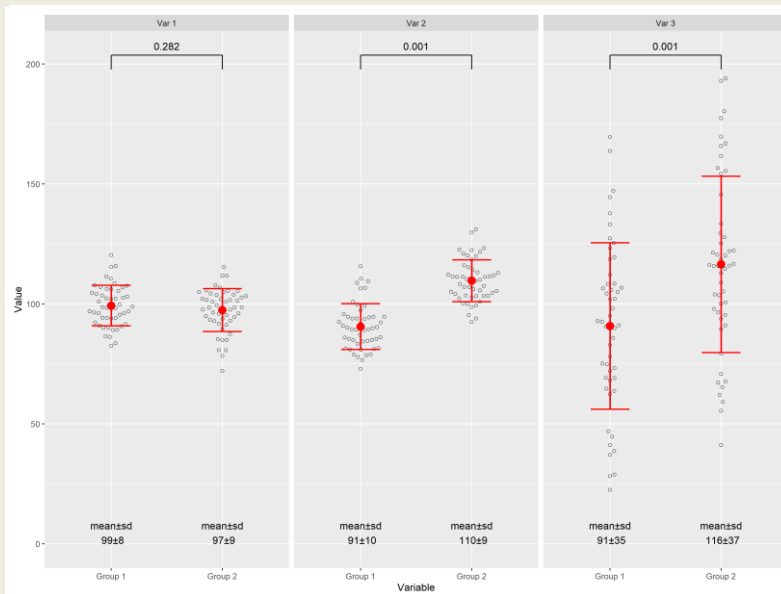
$$p \pm Z * (p * (1-p) / n)^{.5}$$

FDP: 6% (4.5 .. 7.5, n=1000) (4 .. 8, n=500)

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Mean difference

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Comparing means: t-Test

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- **Assumptions: Continuous data with Normal distribution**
- **1 or 2 (independent or dependent) samples**
- **Underlying logic 1 sample-test:**
 - The empirical arithmetic mean $\bar{X}$ of a random variable with true mean μ is a random variable with Normal distribution
 - The standardized difference between expected (true) mean and empirical mean is following a 't' distribution with n-1 degrees of freedom:

$$T = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} = \frac{\bar{X} - \mu}{SEM}$$

- T increases with n and difference but decreases with S
- If this test-value is unlikely, i.e. big: $|t| > t(1 - \frac{\alpha}{2}, n - 1)$
your data seems to differ in mean from your expectation / Null hypothesis

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Comparing means: t-Test

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- Underlying logic 2 sample-test:
 - The difference between two sample means μ_x and μ_y is a random variable $D = \bar{x} - \bar{y}$
 - The hypothesis that $\mu_x = \mu_y$ can be transformed into $D=0$
 - This is the known 1-sample case!
 - Expected mean difference is 0

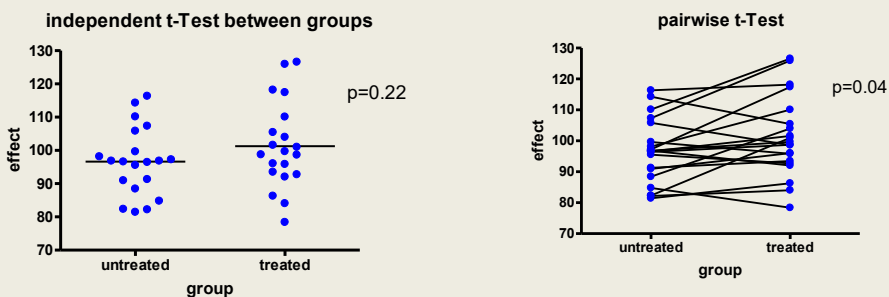
$$T = \sqrt{n} \frac{D}{S}$$

- Somewhat more complicated, if $SD_x \neq SD_y$, S has to be weighted
- For pairwise t-test: s is SD of within-pair differences

53

Independent Samples vs. Paired Test

54



- t-Test with independent groups: Different subjects, smaller power due to variation within groups
- t-Test with dependent groups: Comparison of two values in identical subjects

54

t-Test in R

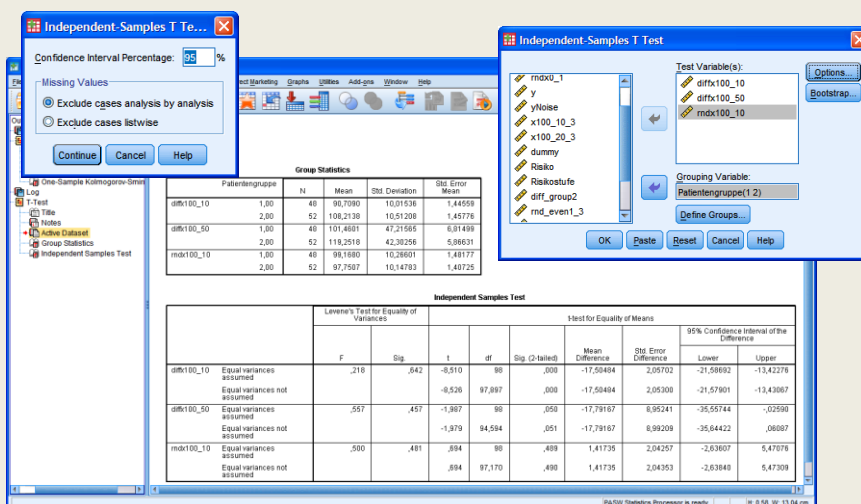
55

- `vartest_out<-var.test(groesse~geschlecht)`
- `t.test(groesse~geschlecht,
var.equal=vartest_out$p.value>.05)`
- `t.test(blutdruck_systol_einschluss,
blutdruck_systol_abschluss,
alternative='greater', # x>y
paired=T) #pairwise t-test, within subject`

55

Missing Data and Many Variables

56



The screenshot displays the SPSS 'Independent-Samples T Test' dialog box and the resulting output table. The dialog box shows the 'Test Variable(s)' as 'diffx100_10', 'diffx100_50', and 'mdx100_10'. The 'Grouping Variable' is 'Patientengruppe(1,2)'. The output table shows the results of the t-test for each variable, including the mean difference, standard error, and 95% confidence interval.

Test Variable(s)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference
diffx100_10	-17,58484	2,05702	-21,58692 -13,42276
diffx100_50	-17,78167	6,95241	-35,55744 -2,00590
mdx100_10	1,41735	2,04353	-2,63848 5,47318

56

Two-tailed vs. One-tailed p Value

57

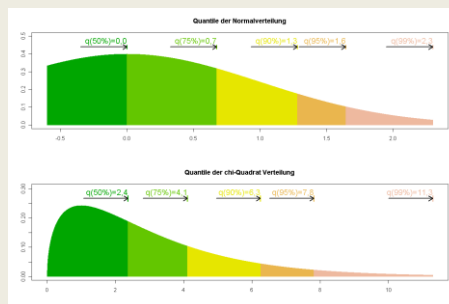
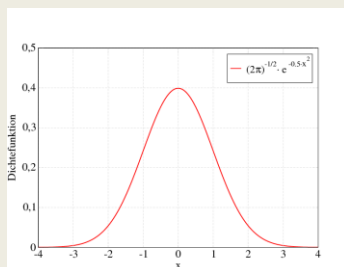
- General question of a statistical test: Assuming the Null hypothesis is true, what is the chance that randomly selected samples would have means as far apart as (or further than) you observed in your experiment with either group having the larger mean?
- Sometimes previous data, physical limitations or common sense predict the direction of the difference or just 1 direction is of interest
- In these cases, a one-tailed test MAY be appropriate
- One-tailed (or 1-sided) test may be selected in options or derived by $p/2$

57

Two-tailed vs. One-tailed p Value

58

- Most statistically relevant distributions are symmetric (Gaussian Normal, t ...), probability thresholds are related to both ends (positive/negative difference)
- Beware, some distributions aren't (e.g. χ^2)!



58

Two-tailed vs. One-tailed p Value

59

- You should only choose one-tailed P values when
 - You predicted direction of difference BEFORE data collection
 - AND
 - You would attribute any difference in the opposite direction – even if it was LARGE – to chance
- Arguments for two-tailed P values
 - Relationship between P values and confidence intervals is easier to understand with two-tailed test
 - When comparing 3 or more groups, test becomes multi-tailed, the two-tailed test is conceptually closer to this generalized case

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Dilemma of One-tailed Test

60

- What do you do if the 'wrong' group ends up with the larger mean?
- It would be statistically correct to conclude that this difference is a chance result, even when it is huge
- Yet there may be a flaw in your theory and you are about to throw out valuable evidence
- Don't give in to the temptation of just flipping your expectation/hypothesis and do a one-tailed test in the other direction
- Switch to two-tailed?

60

Some Other Relevant Tests

61

- Mann-Whitney-U-Test
 - Common nonparametric test for two independent samples; values from both groups are combined and ordered, mean ranks are compared between groups
- Friedman Test
 - Non-parametric version of repeated-measure design
 - Test, if k Variables measured in n subjects are all from the same population / distribution
 - For each case the k variables will be assigned ranks
 - Ranks are compared between variables

61

U-test

62

- Arrange all the observations into a single ranked series. That is, rank all the observations without regard to which sample they are in.
- Add up the ranks in sample 1. The sum of ranks in sample 2 follows by calculation, since the sum of all the ranks equals $N(N + 1)/2$ where N is the total number of observations.
- "U" is then given by the minimum of:

$$U = n_1 * n_2 + R_1 - n_1 (n_1 + 1) / 2 \text{ (for } R_1 \text{ and } R_2 \text{)}$$

where n_1 and n_2 are the sample sizes,

R_1 is the sum of ranks in group 1.

62

U-Test in R

63

- `wilcox.test(x, y = NULL, alternative = c("two.sided", "less", "greater"), mu = 0, paired = FALSE, exact = NULL, correct = TRUE, conf.int = FALSE, conf.level = 0.95, ...)`
- `wilcox.test(NYHA_Einschluss~geschlecht,exact=F)`
- Wilcoxon rank sum test with continuity correction

data: NYHA_Einschluss by geschlecht

W = 12.5, p-value = 0.711

alternative hypothesis: true location shift is not equal to 0

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Covariance / Correlation

64

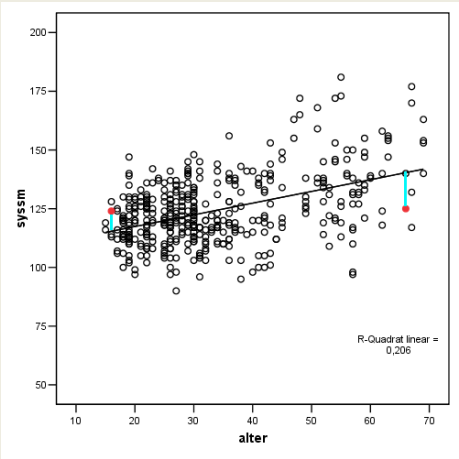
- Covariance is a measure of how much two random variables vary *together* (variance measures how much a single variable varies)
 - If one variable is above its expected value while the other variable tends to be above *its* expected value too, then the covariance between the two variables will be positive.
 - If one of them is above its expected value while the other variable tends to be *below* its expected value, then the covariance between the two variables will be negative
 - Defined as $\text{Cov}(x,y) = \text{Sum } (x - \text{mean}_x)(y - \text{mean}_y) / (n - 1)$
 - Value is dependent on scaling of x,y
- Correlation
 - Standardized covariance $\{-1 \dots 1\}$
 - Defined as $r = \text{Cov}(x,y) / \text{SD}_x * \text{SD}_y$
- Does NOT imply causation
 - Storks $\rightarrow$ birth rate; piracy $\rightarrow$ CO₂

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Linear Regression

65

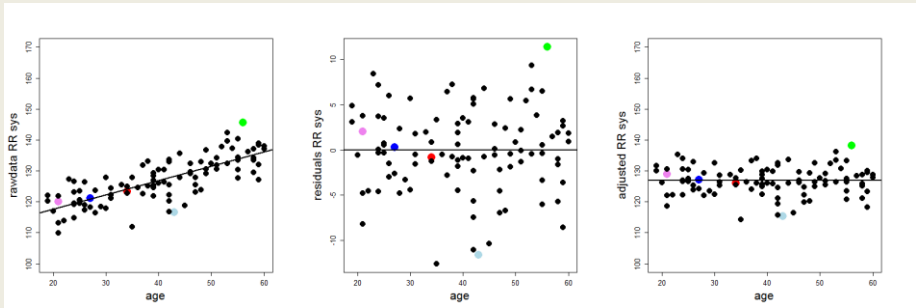
- Relation between two or more quantitative traits
- $y=a+bx$ {sysm=107+0.5*age}
- Reports strength (slope of regression line) and direction
- Can be used to analyze relations, to predict, to estimate missing values
- Frequently applied to 'correct' or adjust values (e.g. estimate age- and sex-independent values) by calculation of residuals (difference between actual value and prediction on regression line)



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Adjusted Values

66



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Regression in R

67

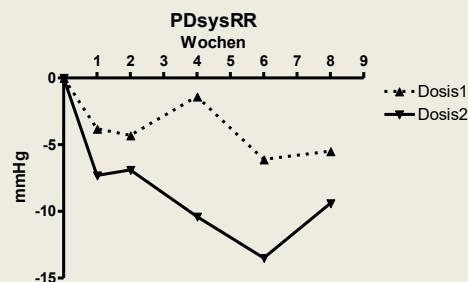
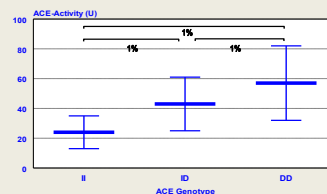
- `plot(y=Size,x=Age) #creates scatterplot`
- `lmOut<-lm(Size~Age)`
- `abline(lmOut) #plots regression line`
- `lmOut$residuals # no age effect, basis for adjusted values`
- Prediction:
 - ▶ `predicted<-data.frame(Age=10:90) #Outside data range!`
 - ▶ `predicted$Size<-predict(lmOut,newdata=predicted)`
 - ▶ `plot(y=predicted$Size,x=predicted$Alter)`

67

Analysis of Variance ANOVA

68

- Test for MEAN differences in more than 2 groups
- Groups may be independent (e.g. 3 Genotypes, treatment groups...)
- Data could be obtained in one group (repeated measures, e.g. before and during therapy)
- Complex extensions for simultaneous comparisons of inter- and intra-individual differences



68

Multiple Testing and p

69

- If you repeat your trial, what is the chance at least 1 result is a false positive?
- For 2 comparisons: $(0.05+0.95)^2$ tells us that
 - p for both comparisons false positive = 0.0025
 - p for at least 1 comparison false positive = 0.095
 - p for no false positive = 0.9025
- For 3 comparisons, $p(\text{at least 1 false positive})=0.14$
- Pulpo Paul and the 8 correct soccer predictions
 - Probability: 0.4% for 1 animal
 - 3,8% for 10 animals and 32% for 100 animals worldwide
 - It just like winning the jackpot: extremely small probability but large number of gamblers!

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Multiple Modelling

70



70

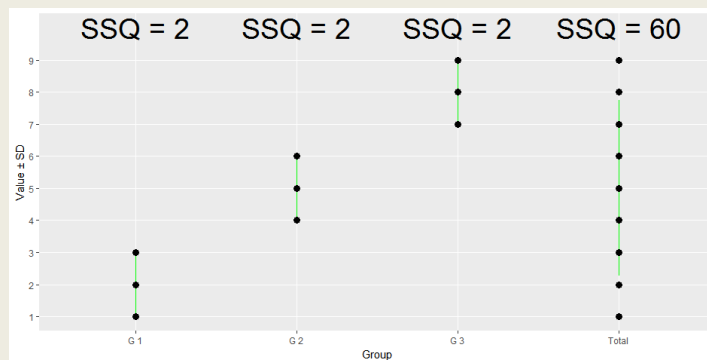
71



71

Variance-Decomposition

72

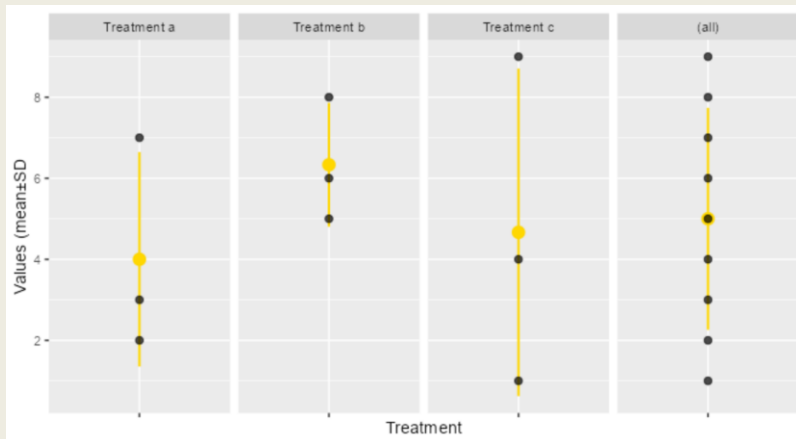


- $SSQ = \text{Sum of Squared Differences}$
($\text{Varianz} = SSQ/n-1$)
- $SSQ_{\text{total}} = SSQ_{\text{within}} + SSQ_{\text{between}}$
- $SSQ_{\text{between}} = 60 - (2+2+2) = 54 \neq 0$

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Variance-Decomposition

73



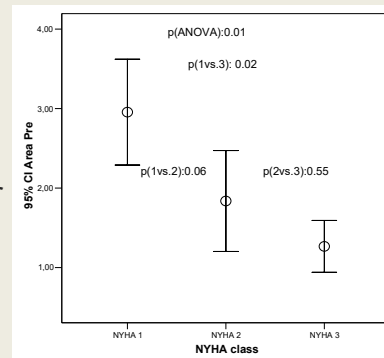
■ $SSQ_{\text{between}} = 60 - (14.0 + 4.7 + 32.7) = 8.6 \approx 0$

73

Post-hoc Comparisons

74

- ANOVA computes p-value for mean difference(s) across all groups, a deviation in 1 out of n groups is sufficient
- Post-hoc analysis computes pairwise comparisons, corrected for multiple testing
- Most often Bonferroni ($p \times \text{number of comparisons}$) or Scheffe-test, various other...



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SPSS post-hoc Tests: Overview

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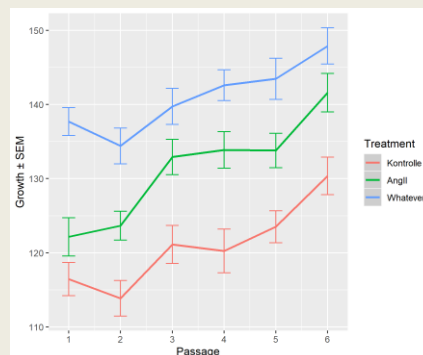
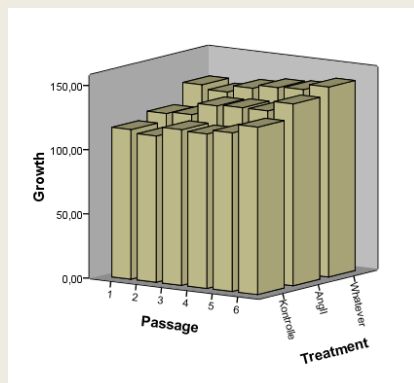
- LSD. Uses t tests to perform all pairwise comparisons between group means. No adjustment is made to the error rate for multiple comparisons.
- Bonferroni. Uses t tests to perform pairwise comparisons between group means, but controls overall error rate by setting the error rate for each test to the experimentwise error rate divided by the total number of tests...
- Sidak. Pairwise multiple comparison test based on a t statistic. Sidak adjusts the significance level for multiple comparisons and provides tighter bounds than Bonferroni.
- Scheffe. Performs simultaneous joint pairwise comparisons for all possible pairwise combinations of means. Uses the F sampling distribution. Can be used to examine all possible linear combinations of group means, not just pairwise comparisons.
- ...
- Waller-Duncan. Multiple comparison test based on a t statistic; uses a Bayesian approach.
- **Dunnett**. Pairwise multiple comparison t test that compares a set of treatments against a single control mean. The last category is the default control category. Alternatively, you can choose the first category ...
- For unequal variances:
 - Tamhane's T2. Conservative pairwise comparisons test based on a t test.
 - • Dunnett's T3. Pairwise comparison test based on the Studentized maximum modulus.
 - • Games-Howell. Pairwise comparison test that is sometimes liberal.
 - • Dunnett's C. Pairwise comparison test based on the Studentized range.

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Univariate ANOVA: Multiple Factors

76

- In your experiment, there is more than 1 source of variation
- Not all potential influences have to be of interest (e.g. variation between hospitals/labs or passages of a cell culture)
- Different groups of a factor do not have to be ordered



76

Types of Sum of Squares

77

- *Type I, “sequential” sum of squares:*
 - $SS(A)$; $SS(B | A)$; $SS(AB | B, A)$.
 - Results may vary depending on order of variables.
- *Type II:*
 - $SS(A | B)$; $SS(B | A)$
 - Every variable is tested as LAST
 - no significant interaction assumed
 - Greater statistical power as Type III
- *Type III:*
 - $SS(A | B, AB)$; $SS(B | A, AB)$;
 - Main effects may be difficult to interpret in the presence of interactions!

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Anova in R

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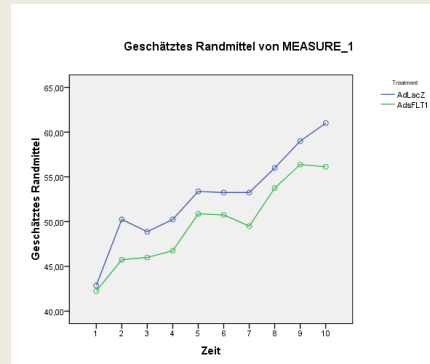
- Anova and linear Regression are not different functions in R
- Function `lm()` can handle quantitative predictors (Regression) and factorial predictors (ANOVA)
- Type of predictor (numeric vs. factor/character) defines type of model
- `lmOut<-lm(growth~as.factor(passage)*condition, data=rawdata)`
- Function `aov()` is a „wrapper“ for `lm()`, reporting in the traditional language of ANOVA
- `Anova(lmOut)` and `anova(lmOut)` compute p-values, using type 2/3 or type 1(order matters!)

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Repeated Measures ANOVA

79

- Two or more groups with more than one observation per individual
- Different measures or different time points
- Test for within-subject effects (e.g. time)
- Test for between-subject effects (e.g. treatment)
- Test for interaction



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Repeated measures ANOVA in R

80

- Different ways to achieve (more or less) the same:
 - ▶ `summary(aov(growth~treatment*time+Error(TierNr), data=rawdataLong))`
 - ▶ `ezANOVA(data=rawdataLong, dv=growth, wid=TierNr, type=1, between=treatment, within=time)`
 - ▶ `anova(lme(fixed=growth~treatment*time, random=~1|TierNr, data=rawdataLong))`

80

General Linear Models

81

- ANOVA and Linear Regression are univariable Versions of LM
- GLM takes 2 or more independent variables into account, but still has 1 measured dependent variable with residuals expected to follow a Normal distribution
- Problems due to multivariable approach:
 - Collinearity (Correlation between predictor variables)
 - Early variables explain some SSQ of later variables
 - Order becomes important
 - Interaction
 - Non-linearity of some predictors
 - May appear/disappear after earlier variables

81

Selection of Predictor Variables

82

- Models should be parsimonious, e.g. contain a reasonably small number of predictors
- Selection may be
 - Backward: Start with full model and stepwise drop the least significant effect
 - Forward: Start with most significant parameter and stepwise add another single parameters with highest significance
 - Best Subset: Test ALL possible combinations
- Argument for smaller p: Multiple testing
- Argument for bigger p: Include all relevant variables

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Generalized Linear Models

83

- In LM/GLM, residuals have to follow a Normal distribution, so usually we expect the dependent variable to be a measured value
- For count data with Poisson distribution and factorial data with binomial distribution, output from linear models and dependent variable will be connected by a 'link' function
- As there is no numeric solution for this kind of models (as in LM/GLM), iterative solutions to find Maximum Likelihood are employed
- Logistic regression is a common term for yes/no outcome variables

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Generalized Linear Models

84

- Identity link
 - Normal distribution

$$g(y) = y = \mathbf{X}\beta + \epsilon$$

- Log link
 - Poisson distribution

$$g(y) = \ln(y/(1-y))$$

$$y = e^{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots} + \epsilon$$

- Logit link
 - Binomial distribution

$$y = \frac{e^{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots}}{1 + e^{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots}} + \epsilon$$

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Logistic Regression

85

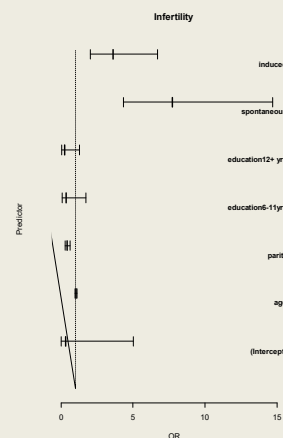
- Binary outcome yes/no
- Odds ratio (OR):
 - Number of yes/number of no, 2:3
 - Differs from probability number of yes/total number 2/5
- Link function logit: $\text{logit}(y) = \ln(y/(1-y))$
- Regression: $\text{logit}(y) = mx + b$
- Prediction of y:
$$y = \frac{e^{mx+b}}{1+e^{mx+b}}$$
- R function: `glm()`
- Prediction: `predict(..., type=response)`

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Odds Ratios / CIs

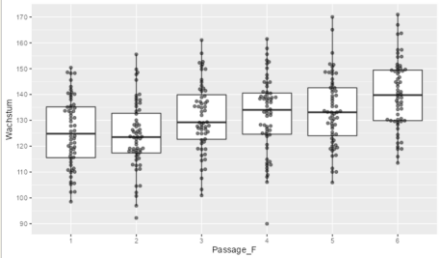
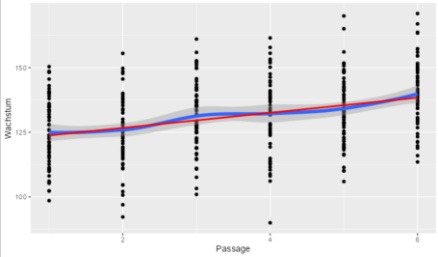
86

- OR: Odds in group 1 (e.g. male)/ odds in group 2 (female), in case of quantitative predictor odds/odds@-1unit
- ORs are depending on the scale of the predictor, normalization may be useful
- CIs are asymmetric, as they are based on exponential transformation



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Regression vs. ANOVA





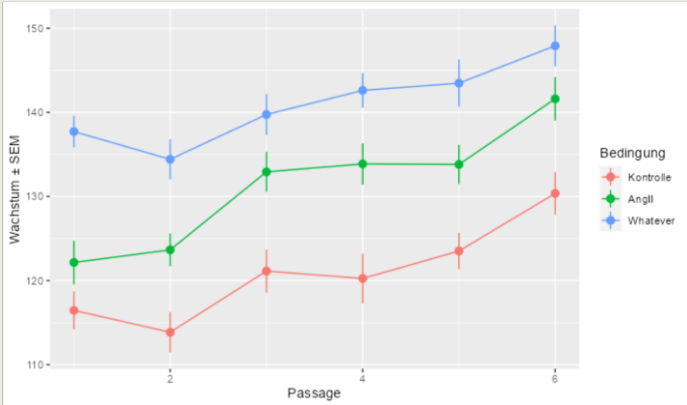
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Linear Models

- Regression and/or ANOVA





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Linear Mixed Models

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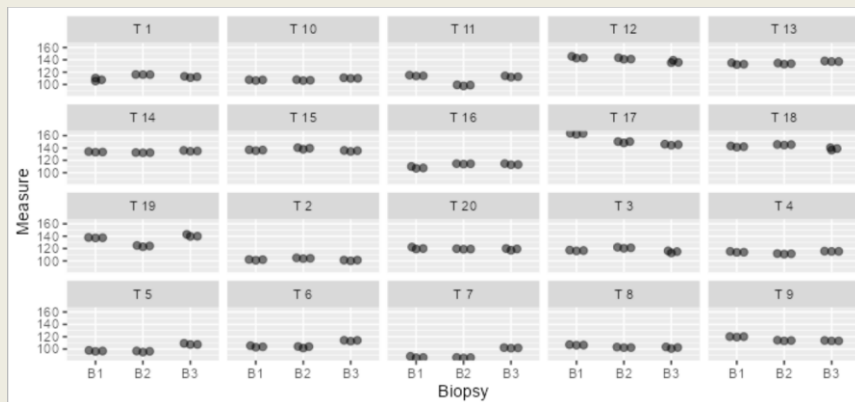
- Comparable to linear models, but with distinction between independent variables:
 - Factors of interest can be nested, within subgroups effects are correlated (repeated measures with subject, technical replicates...)
 - Fixed effects:
 - That is what we want to know and put a p-value to
 - Effect is assumed to be constant / valid for the population / useful for prediction
 - Random effects:
 - Effectsize of no real interest, valid for the sample / group at hand
 - Only variance is estimated and taken out, for better estimation of fixed effects

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Nested Data

90

- Replicate within Biopsy within Animal

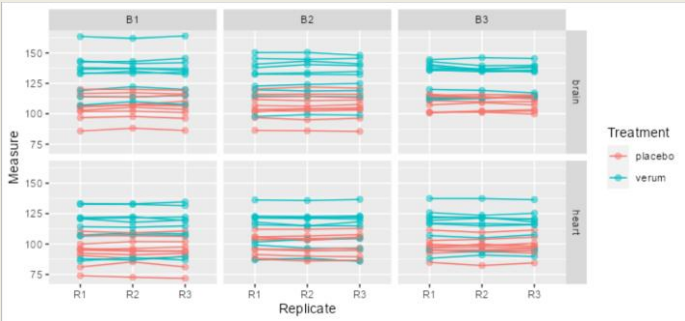


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Within-/Between-Subject

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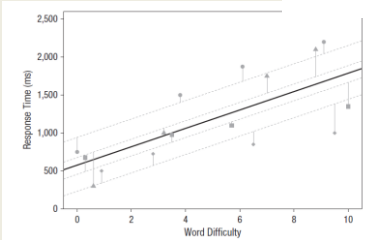
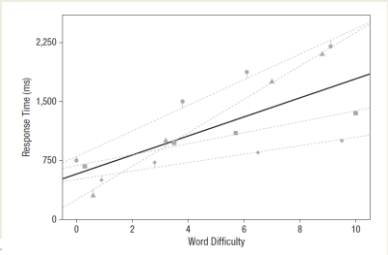
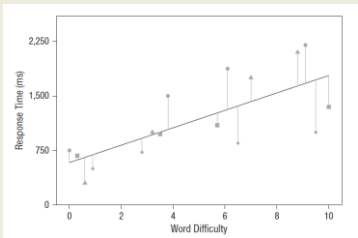
- Within-subject: Replicate/Biopsy/Organ
- Between-subject: Treatment



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Fixed / random intercept / random slope

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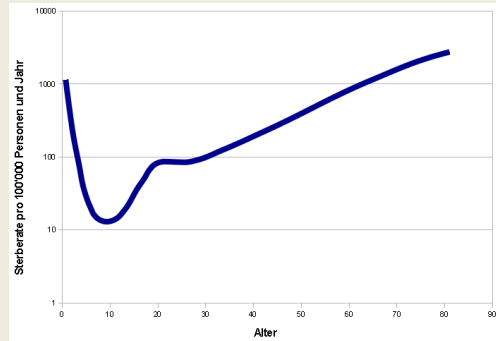


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Life expectancy

93

- Mean life expectancy
 - Males: 76,9 years
 - Females: 82,3 years
- This number is valid for new borns only, once you survived early infancy, you have a fair chance of getting older (especially if you are older than your mean life expectancy, e.g. 85)
- Beware: Bishops have higher life expectancy than mechanics, but not due to life style but simply by having survived into older age already, you may be a mechanic by 20 but a bishop probably not before 50 or so



Age	0	20	40	60	80
Life expectancy	77	77.5	78	80.5	87.5

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Evidence?

94

- Is study design convincing?
- Is the sample representative for my own question?
- How big is the sample (especially relevant when result is negative)?
- How big is the effect (independent of p!), any clinical relevance?
- Large sample + small effect + significance: Interesting but irrelevant?!
- Small sample + relevant effect – significance: Hypothesis still worthwhile testing, larger sample?!
- How many hypotheses were tested simultaneously, adjusted or nominal p values reported? (rule of thumb: $p \times \text{number of tests}$)

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Trust abstract?

95

- Can the Accuracy of Abstracts Be Improved by Providing Specific Instructions? A Randomized Controlled Trial
- [Roy M. Pitkin, MD; Mary Ann Branagan](#)
- JAMA. 1998;280:267-269.

Abstract Defects in Published Articles

Journal	Defective Abstracts, No. (%)	Types of Defects in Abstract			
		Inconsistency	Not in Body	Both	Conclusion
American Journal of Obstetrics and Gynecology	19/36 (53)	11	5	2	1
Pediatrics	13/20 (65)	5	4	4	0
JAMA	7/14 (50)	3	1	3	0
The New England Journal of Medicine	3/11 (27)	1	2	0	0

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Software?

96

- Big packages: SPSS, SAS, Stata, Statistica
 - Rather expensive (if not available via campus-licence)
 - Cover most typical applications
 - Allow the user to make stupid mistakes and run complex analyses without the slightest idea by offering simple menus
- Balance between stats and graphics: GraphPad Prism
- Free software
 - PSPP (a free SPSS clone)
 - R (statistical scripting language)**
 - Flexible but demanding, avoid if you hate computers

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